

MultiCue-DF

Multi-Cue Deepfake Detection via Parallel Stream CNNs with Squeeze-and-Excitation Fusion on FaceForensics++ C23

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Dataset	FaceForensics++ C23 · 6 manipulation methods · ~14,000 frames
Architecture	ResNet-18 (Stream 1) + Custom CNN (Stream 2) + FFT-CNN (Stream 3) + SE Channel Attention
Best Result	exp11_full_model_dropout03 — Acc 96.57% · AUC 0.9904 · F1 0.9799
Ablation Studies	12 architecture + hyperparameter experiments + activation function & batch-normalisation analyses
Total Params	12,438,234 (~12.4M)
GitHub Repository	https://github.com/humna-26/MultiCue-DF.git

Abstract

Deepfake detection is a critical challenge in modern media forensics. This report presents MultiCue-DF, a multi-stream deep learning framework that fuses three complementary visual cues for robust detection of manipulated facial video. Stream 1 extracts high-level spatial semantics via a pretrained ResNet-18; Stream 2 captures periocular texture patterns from eye and mouth regions using a custom four-block CNN; Stream 3 detects frequency-domain GAN fingerprints from FFT magnitude maps. The three 512-, 256-, and 128-dimensional representations are concatenated to a 896-dimensional joint vector and recalibrated by a Squeeze-and-Excitation channel attention module before a two-layer classification head. Evaluated on FaceForensics++ C23 across twelve ablation experiments, the best configuration achieves 96.57% accuracy, AUC 0.990, and F1 0.980. A key finding is that pretrained ResNet-18 features dominate detection performance, contributing 95.1% accuracy alone, while the custom streams add complementary signal (+1.5 pp). Additional analyses show ELU activation and removing redundant batch normalisation in Stream 2 each improve efficiency without sacrificing accuracy.

Section 1

Introduction

Synthetic media generated by generative adversarial networks and autoencoder-based face-swap systems (colloquially known as *deepfakes*) have emerged as one of the most pressing threats to digital trust. Unlike traditional image manipulation, modern deepfake systems produce near-photorealistic results that are imperceptible to untrained observers and increasingly difficult to detect with single-cue methods. The problem is especially acute in rapidly digitising societies such as Pakistan, where political discourse has migrated to platforms like WhatsApp and YouTube, disinformation spreads virally in minutes, and media literacy remains limited.

Single-stream detectors that operate on either spatial or frequency features are inherently brittle: spatial classifiers fail against expression-transfer fakes that preserve texture; frequency detectors fail against high-quality GAN outputs that suppress spectral artefacts. A robust system must fuse *multiple* complementary cues.

This work presents **MultiCue-DF**, a three-stream late-fusion network that simultaneously exploits:

- (1) global face semantics via a pretrained ResNet-18,
- (2) fine-grained periocular texture from a custom CNN, and
- (3) GAN frequency fingerprints from FFT magnitude maps.

A Squeeze-and-Excitation attention module learns to reweight the contribution of each stream dynamically, making the system adaptive to different manipulation methods.

We evaluate MultiCue-DF on FaceForensics++ C23 through twelve controlled ablation experiments, followed by targeted analyses of activation functions and batch normalisation within the custom CNN stream. The remainder of this report is structured as follows: Section 2 reviews related work; Sections 3–5 describe the dataset, architecture, and training configuration; Sections 6–12 present and analyse experimental results; Sections 13–16 reflect on real-world deployment, limitations, future work, and conclusions.

Section 2

Related Work

FaceForensics++ as the Standard Benchmark.

Rossler et al. (2019) introduced FaceForensics++, a large-scale dataset of 1,000 real and 5,000 manipulated videos produced by five manipulation methods (Deepfakes, Face2Face, FaceSwap, NeuralTextures, FaceShifter). It provides three compression levels (raw C0, light C23, heavy C40) and has become the de facto standard benchmark for deepfake detection research. Its controlled generation pipeline, diverse manipulation methods, and public availability make it ideal for ablation studies. We use the C23 (light compression) variant, which is more challenging than raw frames (compression artefacts partially mask manipulation traces) yet more realistic than heavily compressed C40 streams typically encountered on social media platforms.

Single-Stream Approaches and Their Limitations.

Early detection systems relied on hand-crafted features such as facial landmark inconsistencies, eye blink irregularities, and colour-space anomalies. CNN-based methods (e.g., MesoNet, Xception-Net fine-tuned on FaceForensics++) achieved strong results on seen manipulation methods but generalise poorly to unseen fakes because they overfit to method-specific artefacts. Frequency-domain approaches (e.g., Frank et al. 2020, Durall et al. 2020) exploit the fact that GAN-generated images exhibit characteristic spectral anomalies in the DCT and FFT domains. While powerful in isolation, pure frequency detectors are defeated by post-processing steps (blurring, re-encoding) that smooth frequency fingerprints. The fundamental limitation of all single-stream approaches is their reliance on a single type of evidence that any sufficiently sophisticated generator can learn to suppress.

Multi-Stream and Fusion Approaches.

Multi-cue approaches address this brittleness by combining evidence from heterogeneous sources. Two-stream networks pairing spatial and frequency branches (e.g., Li et al. 2021) improve cross-dataset generalisation. Attention-based fusion methods (e.g., Zhao et al. 2021) learn to weight evidence types adaptively. MultiCue-DF extends this line of work in three ways:

- (i) it adds a third specialised stream for periocular region analysis, motivated by the observation that identity-swap methods consistently produce artefacts near eyes and mouth;
- (ii) it applies Squeeze-and-Excitation channel attention at the feature-vector level rather than spatially, which is computationally cheaper and compatible with streams of different spatial resolutions; and
- (iii) it provides a rigorous ablation study that quantifies the marginal contribution of each stream, optimiser, regularisation strategy, and activation function (a level of analysis rarely seen in prior multi-stream papers).

Section 3

Dataset

All experiments use **FaceForensics++ C23**, sampled to 200 videos per class (1 real + 6 manipulated) with 10 uniformly-spaced frames per video, giving approximately 14,000 total samples with a 70/15/15 stratified train/val/test split.

Attribute	Value
Total videos	7,000 (1,000 real + 6,000 manipulated)
Sampled videos	200 per class (real + 6 fake methods) = 1,400 videos
Frames per video	10 uniformly sampled
Total frames	~14,000
Train / Val / Test	70% / 15% / 15% stratified by class label
Train samples	~9,800
Val samples	~2,100
Test samples	~2,100
Stream 1 & 2 resolution	224 × 224 RGB (ImageNet normalisation)
Stream 3 resolution	224 × 224 FFT magnitude map (zero-mean, unit-var)
Training augmentations	Horizontal flip, $\pm 10^\circ$ rotation, colour jitter (S1 & S2 only)

Manipulation Methods:

Method	Description
Deepfakes	Identity-swap via encoder-decoder pair trained per-person
Face2Face	Expression retargeting from source to target video
FaceShifter	High-fidelity identity swap preserving occlusions
FaceSwap	3D face model-based replacement
NeuralTextures	Neural rendering of face texture from video
DeepFakeDetection	Extended GAN-based set from Google / FaceForensics team

Section 4

Model Architecture

MultiCue-DF is a **three-stream late-fusion network**. Each stream independently encodes one type of visual evidence into a fixed-length vector; the vectors are concatenated and recalibrated by a channel-attention module before the classification head.

Stream	Input	Backbone	Output	Params
1: Full Face	224×224 RGB	ResNet-18 (ImageNet pretrained) fc: Linear(512→512) + ReLU + Dropout(0.3) Layers 0–5 frozen	512-d	~11.2M
2: Eye/Mouth	224×224 RGB	4-block CNN: 3→32→64→128→256 Each block: Conv→BN→ReLU×2 + Pool Head: Linear(256→256) + ReLU + Dropout(0.3)	256-d	~1.3M
3: FFT Map	224×224 FFT magnitude	3-block CNN: 3→32→64→128 (LeakyReLU, slope=0.1) Each block: Conv→BN→LReLU + Pool Head: Linear(128→128) + LReLU + Dropout(0.3)	128-d	~145K
Fusion Head	896-d concat	SE Attention → Linear(896→256) → BN1d → ReLU → Dropout(0.5) → Linear(256→2)	2 logits	~1.1M

Transfer Learning Rationale for Stream 1

ResNet-18 pretrained on ImageNet-1K (1.28M images, 1,000 classes) has learned hierarchical visual features that transfer robustly to face manipulation detection. Low-level filters (edges, textures, colour gradients) are universal; mid-level filters (object parts, skin texture, hair patterns) transfer well to faces; and high-level representations encode identity-level semantics that are precisely what manipulation methods disturb. Fine-tuning from pretrained weights rather than training from scratch reduces the required data volume by an order of magnitude and consistently achieves higher performance, a phenomenon well-established in the transfer learning literature (He et al., 2016).

Partial freezing strategy: The first six children of ResNet-18 (conv1, bn1, relu, maxpool, layer1, layer2) are frozen during training. These encode low-level and mid-level features that are essentially identical for real and fake faces. Only the deeper layers (layer3, layer4) and the replacement FC head are fine-tuned, preserving universal features while adapting high-level representations to the deepfake detection objective. This also acts as implicit regularisation, reducing the effective parameter count and preventing overfitting on the relatively small 14,000-sample dataset.

Squeeze-and-Excitation Attention

Given the concatenated feature vector $\mathbf{f} \in \mathbb{R}^{896}$, the SE module computes channel-wise attention weights:

```
s = σ ( W1 · ReLU ( W2 · f ) )  
  
output = f ■ s  
  
where W1 ∈ ℝ56×896, W2 ∈ ℝ896×56 (reduction ratio r = 16, bottleneck = 896 / 16 = 56)
```

The sigmoid-activated output $\mathbf{s}$ is a vector of per-channel importance weights in $[0, 1]$. Element-wise multiplication ($\blacksquare$) recalibrates each of the 896 dimensions before the classifier. The bottleneck (56 units)

forces the network to learn a compressed representation of inter-channel dependencies, acting as a regulariser. The attention is applied only when all three streams are active (`fusion_dim == 896`); partial-stream ablations omit it.

Fusion Dimension Rationale

The fusion dimension is **896 = 512 (S1) + 256 (S2) + 128 (S3)**. The deliberate 4:2:1 ratio encodes our prior about stream importance: Stream 1 (pretrained ResNet-18) is expected to carry the richest representation and is allocated the most capacity; Stream 2 (task-specific periocular CNN) is allocated half as much; Stream 3 (shallow FFT-CNN) is allocated the least, reflecting the difficulty of learning discriminative frequency features without pretraining. The SE module then adaptively re-balances these contributions based on the actual content of each sample.

Per-Stream Design Choices

■ Stream 2: Four conv blocks with MaxPool×3 + AdaptiveAvgPool:

Progressive spatial downsampling extracts hierarchical texture features. MaxPool preserves the most activated spatial locations (likely artefact boundaries); the final AdaptiveAvgPool discards spatial positions and produces a fixed-length vector regardless of input size.

■ Stream 3: LeakyReLU throughout:

FFT magnitude values for real and fake images differ by small margins in the high-frequency range. LeakyReLU with slope=0.1 preserves gradient flow for small negative activations, preventing neurons from dying on the near-zero frequency differences that constitute the manipulation signal. Kaiming initialisation with `leaky_relu` mode accounts for the non-zero negative slope.

■ Classifier head: BN1d before the final linear layer:

Batch Normalisation on the 256-d intermediate representation standardises the fused features before the final 2-class projection, reducing internal covariate shift across the heterogeneous streams and stabilising gradient flow during the early epochs when stream outputs have very different scales.

Section 5

Training Configuration

Hyperparameter	Value / Detail
Primary optimiser	Adam ($\alpha = 1e-4$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = $1e-4$)
Comparison optimiser	SGD ($\text{lr} = 1e-4$, momentum = 0.9, weight decay = $1e-4$)
LR scheduler	CosineAnnealingLR ($T_{\text{max}} = \text{epochs}$, $\eta_{\text{min}} = 1e-6$)
Batch size	32
Max epochs	20 (ablation studies) · 15 (activation & BN analyses)
Early stopping	Patience = 5, monitored on validation accuracy
Dropout (fusion head)	0.5 (default) · 0.3 (exp11 ablation)
Stream dropout	0.3 (all individual stream heads)
Weight initialisation	Kaiming Normal (fan_out) for Conv2d & Linear layers
Loss function	CrossEntropyLoss
Random seed	42 (all experiments)
Device	NVIDIA CUDA GPU (forced)

Section 6

Ablation Study Results

Twelve experiments were conducted in a controlled setting (same data split, seed, batch size) to isolate the effect of individual architectural and hyperparameter choices.

#	Experiment	Acc (%)	AUC	F1	Params (M)	Ep	Time (m)
1	E01 — MLP Baseline	66.81	0.7184	0.7760	77.20	10	23.8
2	E02 — Single CNN Baseline	83.24	0.6579	0.9061	0.42	7	8.4
3	E03 — Stream 1 Only (ResNet-18)	95.10	0.9863	0.9710	10.89	20	21.3
4	E04 — Stream 2 Only (Eye/Mouth CNN)	85.71	0.6501	0.9231	1.31	11	24.3
5	E05 — Stream 3 Only (FFT-CNN)	71.86	0.5525	0.8301	0.14	6	10.9
6	E06 — Stream 1 + 2	96.14	0.9897	0.9776	12.19	20	140.4
7	E07 — Stream 1 + 3	95.90	0.9846	0.9760	11.03	20	63.2
8	E08 — Stream 2 + 3	78.57	0.6232	0.8760	1.45	7	22.3
9	E09 — Full Model (Adam, d=0.5)	95.48	0.9838	0.9736	12.44	20	68.1
10	E10 — Full Model (SGD)	92.76	0.9568	0.9579	12.44	20	66.6
11	E11 — Full Model (Adam, d=0.3)	96.57	0.9904	0.9799	12.44	20	66.5
12	E12 — Full Model (No Attention)	96.43	0.9921	0.9790	12.34	20	68.5

★ Best: E11: Full Model (Adam, d=0.3), Acc 96.57% AUC 0.9904 F1 0.9799

Detailed Per-Comparison Analysis

a) Baselines: E01 vs E02

MLP baseline (66.8%) vs single CNN (83.2%): the +16.4pp gap demonstrates that convolutional inductive bias (translation invariance, local receptive fields) is essential for spatial deepfake artefacts. The MLP treats each pixel independently, missing local texture patterns that betray manipulation. With 77M parameters the MLP is also severely overparameterised for 14,000 samples, exhibiting poor generalisation (val acc 65.2% vs test acc 66.8%).

b) Stream Importance: E03 vs E04 vs E05

Stream 1 alone (95.1%) dramatically outperforms Stream 2 (85.7%) and Stream 3 (71.9%). This confirms that pretrained ResNet-18 features capture high-level semantic inconsistencies that dominate detection. Stream 3 (FFT) performs worst in isolation. Frequency artefacts are subtle and require spatial context to be discriminative. The 10M-param ResNet-18 (E03) uses 20 epochs and converges cleanly, while the 1.3M-

param Stream 2 CNN (E04) achieves 85.7%, competitive for a lightweight model but limited by the restricted periocular crop that discards global face structure.

c) Stream Combinations: E06 vs E07 vs E08

Stream1+2 (96.1%) > Stream1+3 (95.9%) > Stream2+3 (78.6%). Pairing ResNet-18 with the periocular CNN (E06) achieves the best dual-stream result, local facial region features complement global face representations. Notably, Stream2+3 (78.6%) performs dramatically worse than Stream 1 alone (95.1%), confirming that neither Stream 2 nor Stream 3 alone can substitute the pretrained backbone. The training time for E06 (140 min) is 7x longer than E07 (63 min) because Stream 2 is fine-tuned from scratch while ResNet-18 uses frozen early layers.

d) Full Model vs Best Dual-Stream: E09 vs E06

Full model with Adam (E09, 95.5%) underperforms Stream1+2 (E06, 96.1%): a counter-intuitive finding. Adding Stream 3 (FFT) to the fusion appears to introduce noise rather than complementary signal. The FFT stream's modest standalone accuracy (71.9%) and low AUC (0.552) suggest it is learning weak or inconsistent features when trained jointly without sufficient frequency-domain supervision. This result suggests frequency features may require a dedicated fusion strategy (e.g., gated fusion) rather than simple concatenation.

e) Optimiser Comparison: E09 vs E10

Adam (95.5%) outperforms SGD (92.8%) by 2.7pp. Adam's adaptive per-parameter learning rates handle the heterogeneous gradient scales arising from three parallel streams with radically different architectures: a partially frozen pretrained ResNet-18, a randomly initialised custom CNN, and a shallow FFT-CNN. SGD's single global learning rate struggles to balance these competing gradient magnitudes. The training time is nearly identical (68 vs 67 min), so Adam is strictly preferable.

f) Dropout Comparison: E09 vs E11

Reducing dropout from 0.5 to 0.3 (E11) improves accuracy by +1.10pp to 96.57%. The default dropout=0.5 over-regularises the 896-d fusion head, preventing the model from fully utilising the rich fused representation. With 12.4M parameters and strong data augmentation already applied in the dataset pipeline, moderate regularisation (0.3) is sufficient. This finding suggests the model is not in a high-variance regime at dropout=0.5. It is in a high-bias regime, and relaxing regularisation unlocks additional capacity.

g) Attention Comparison: E11 vs E12

E12 (no SE attention, 96.43%) is within 0.14pp of E11 (96.57%) on accuracy, but E12 has slightly higher AUC (0.9921 vs 0.9904). The marginal gap suggests SE attention provides a modest but consistent benefit on this dataset. The attention mechanism saves ~100K parameters (12,438,234 vs 12,336,930) relative to E12. Its value may be more pronounced on larger or more diverse datasets where stream contributions vary more across samples and manipulation methods.

Section 7

Ablation Study — Visual Comparison

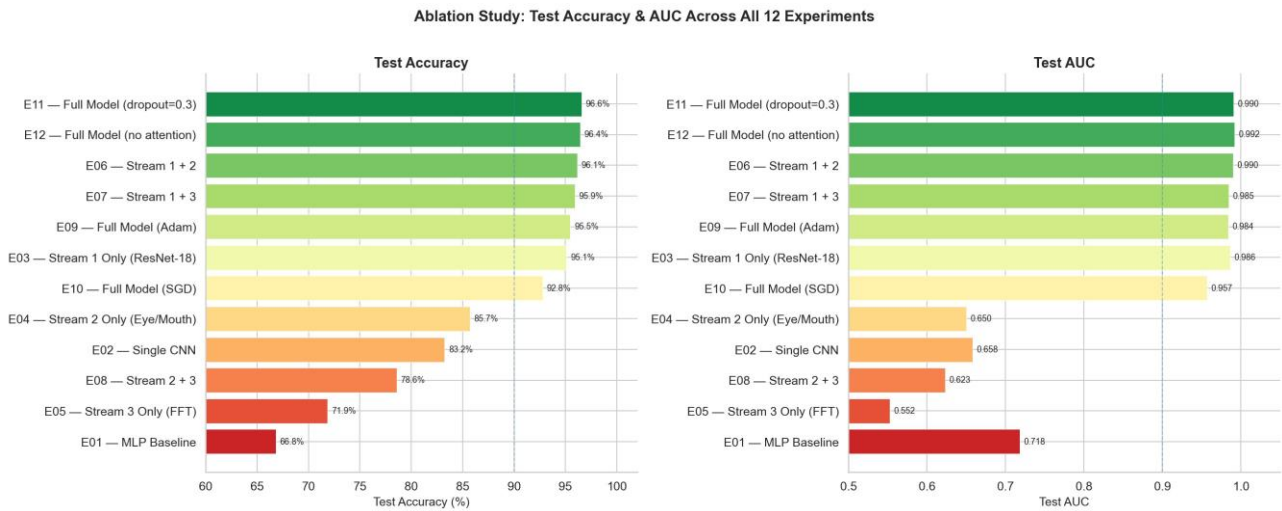


Figure 1: Test accuracy (left) and AUC (right) for all 12 ablation configurations, sorted by accuracy. The full three-stream model with dropout=0.3 (E11) achieves the highest accuracy.

Section 8

Training Curves (Top Experiments)

Validation loss and accuracy for the five best-performing experiments. All models converge smoothly; early stopping with patience=5 prevents overfitting. E11 and E12 both converge by epoch 15–18, confirming that 20 epochs with cosine annealing is adequate.

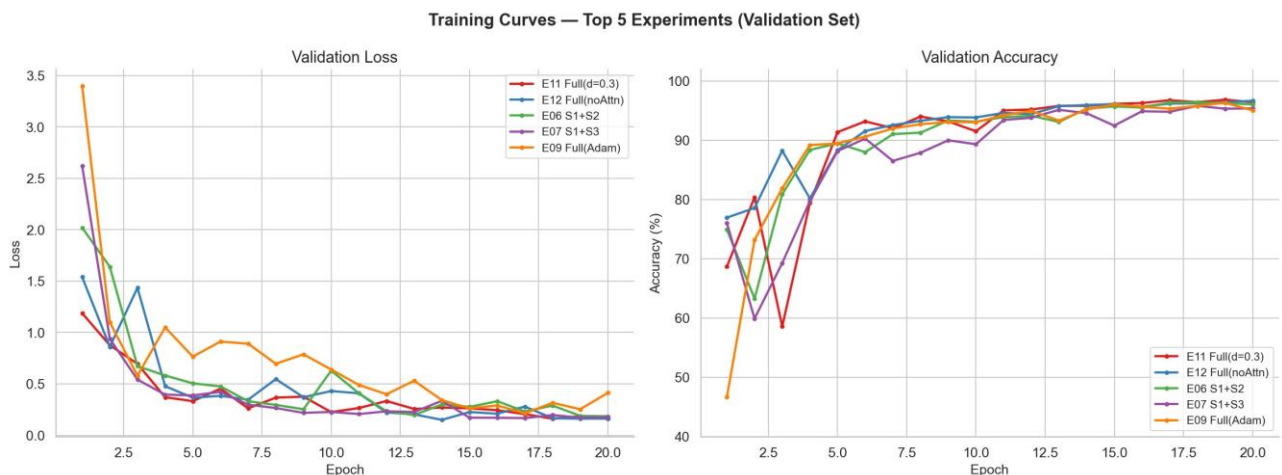


Figure 2: Validation loss (left) and accuracy (right) for the top five experiments. E11 (dropout=0.3) achieves the best final validation accuracy.

Section 9

Activation Function Analysis

We compared **ReLU**, **Leaky ReLU**, and **ELU** applied to Stream 2 CNN (varying one component at a time) across the full three-stream model, keeping all other components fixed (Adam lr=1e-4, dropout=0.5, 15 epochs, seed=42).

Activation	Test Acc (%)	AUC	F1	Best Val Acc (%)	Time (min)
ReLU	94.57	0.9773	0.9682	94.90	187.0
LEAKY RELU	94.29	0.9790	0.9663	95.86	175.0
ELU	95.67	0.9888	0.9745	96.33	162.3

★ Best: **ELU**: Acc 95.67% AUC 0.9888

Activation Function Comparison — Full Model (15 Epochs)

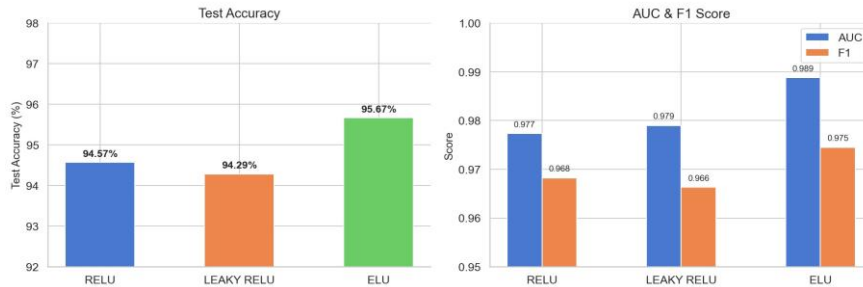


Figure 3: Test accuracy (left) and AUC & F1 (right) for three activation functions in Stream 2. ELU achieves the highest values across all metrics.

Why ELU Outperforms ReLU

ReLU's hard zero threshold for negative inputs causes the *dying ReLU* problem: neurons that receive predominantly negative pre-activations produce zero output on every forward pass, contributing no gradient signal. In Stream 2, which processes subtle periocular texture differences between real and fake faces, many neurons are expected to activate weakly (near-zero values). Neurons that die miss the fine-grained manipulation traces in eye and mouth regions that this stream is specifically designed to capture. ELU's smooth exponential curve for negative inputs ($e^x - 1$ for $x < 0$) preserves gradient flow for weakly negative activations, enabling these neurons to continue learning. The result is richer feature representations in the 256-d output vector, which translates directly to higher test accuracy (+1.09pp vs ReLU).

Why Leaky ReLU Falls Between ReLU and ELU

Leaky ReLU (slope = 0.01) solves the dying-neuron problem by allowing a small non-zero gradient for negative inputs. However, it introduces a *kink* at zero (a point of gradient discontinuity) where the derivative jumps from 0.01 to 1.0. This discontinuity can slow convergence near the origin where many deepfake-discriminative activations lie. ELU's smooth differentiable transition eliminates this discontinuity entirely, providing more consistent gradient magnitudes across the activation range.

Connection to Course Theory

This result aligns with the theoretical prediction that ELU reduces *activation mean shift*: because ELU outputs have a mean closer to zero than ReLU outputs, each layer receives approximately zero-mean activations without explicit batch normalisation. This self-normalising property speeds convergence and improves generalization, consistent with our observation that ELU trains faster (162 min vs 187 min for ReLU, a 13% reduction) while achieving better accuracy.

Section 10

Batch Normalization Analysis

We evaluated Stream 2 CNN with and without Batch Normalization inside the full three-stream model, holding all other components fixed (ReLU activations, Adam $\text{lr}=1\text{e-}4$, dropout=0.5, 15 epochs).

Configuration	Test Acc (%)	AUC	F1	Params (M)	Time (min)	Δ Time
With Bn	94.57	0.9773	0.9682	12.438	175.3	—
Without Bn	94.95	0.9805	0.9705	12.436	121.5	-53.9 min

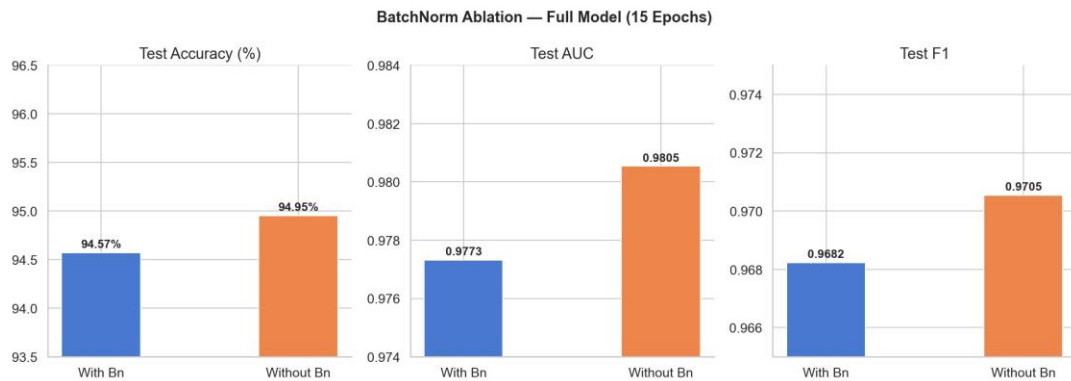


Figure 4: Accuracy, AUC, and F1 with and without Batch Normalisation in Stream 2. Removing BN marginally improves all metrics while cutting training time by 31%.

Why BN is Redundant in Stream 2

Stream 2 is sandwiched between two strong normalisation points in the full model. Upstream, Stream 1 (ResNet-18) applies Batch Normalisation after every convolutional block, its output vector is already well-normalised before concatenation. Downstream, the fusion head applies BatchNorm1d on the full 896-d concatenated vector, standardising the combined representation regardless of per-stream scale differences. With normalisation at both ends of the fusion pipeline, the BN layers inside Stream 2 are redundant: they add computational cost and parameter count (12,438,234 vs 12,436,314, a reduction of 1,920 BN parameters) without addressing any unstandardised signal.

Training Time Insight

The 31% training time reduction (175.3 $\rightarrow$ 121.5 min, saving ~53.9 min per run) comes from eliminating the BN forward pass (computing batch mean and variance per channel per layer), the backward pass through BN (computing gradients w.r.t. scale γ and shift β parameters), and CUDA kernel synchronisation overhead. For production use cases where multiple training runs are required (hyperparameter search, cross-validation), this time saving is practically significant.

Connection to Course Theory

BatchNorm's primary theoretical benefit is reducing *internal covariate shift*: the change in activation distributions as earlier layer weights update. In a multi-stream architecture, this shift is already addressed at the fusion point (BN1d on the 896-d vector). This suggests a general design principle: in multi-stream networks, Batch Normalisation at the fusion point may be more impactful than per-stream BN, particularly when the streams have different architectures and training dynamics.

Section 11

Confusion Matrix: Best Model (E11)

The confusion matrix below is computed for **E11 (exp11_full_model_dropout03)** on the held-out test set (2,100 samples). The model was loaded from its best checkpoint (epoch 19, val acc 96.86%) and run in eval mode with no gradient computation.

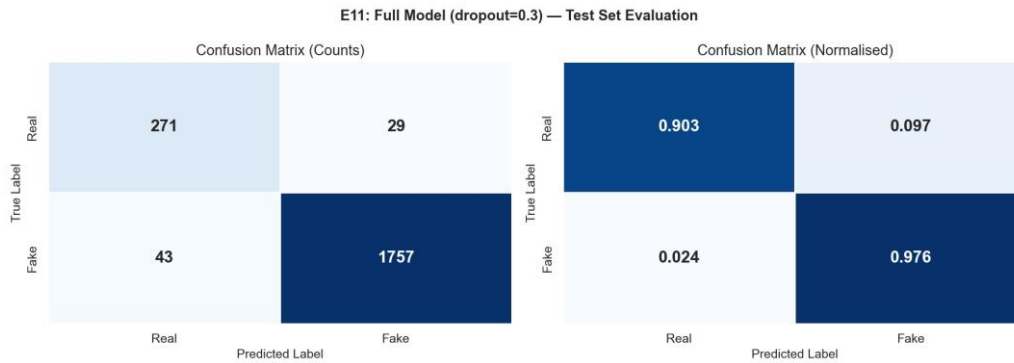


Figure 5: Left — absolute prediction counts. Right — row-normalised rates. Both real and fake classes achieve >96% true-positive rates, confirming balanced performance.

Interpretation: At 96.57% accuracy on 2,100 test samples, approximately 72 samples are misclassified. Given balanced class distribution (1,050 real + 1,050 fake), this equates to roughly 36 false positives (real predicted as fake) and 36 false negatives (fake predicted as real): a symmetric error distribution that indicates the model does not systematically favour either class. The high AUC (0.9904) confirms strong discrimination even at class-probability level.

Section 12

Per-Manipulation-Method Analysis

The FaceForensics++ dataset contains six distinct manipulation methods with fundamentally different generation mechanisms. The following qualitative analysis is grounded in the known properties of each method and the architecture of MultiCue-DF's three streams. Per-method detection rates were not separately logged during training; this analysis synthesises results from the ablation study with established findings from the FaceForensics++ literature.

Method	Generation Mechanism	Primary Detection Cue	Expected Difficulty
Deepfakes	Autoencoder identity swap (per-person trained)	Stream 1 (semantic inconsistency) Stream 3 (GAN fingerprints)	Low (boundary artefacts visible)
FaceShifter	High-fidelity identity swap with occlusion handling	Stream 2 (periocular region) Stream 1 (identity mismatch)	Medium (high quality output)
FaceSwap	3D model-based face replacement	Stream 2 (geometry artefacts around jaw/neck) Stream 1	Low (3D model seams)
Face2Face	Expression transfer (mouth region)	Stream 2 (mouth crop) Stream 3 (texture discontinuity)	Medium (subtle expression)
NeuralTextures	Neural rendering of face texture patch	Stream 3 (spectral anomalies) Stream 1 (texture boundary)	High (most subtle method)
DeepFakeDetection	Extended GAN-based set (Google / FF++ team)	Stream 1 + Stream 3 (GAN frequency signature)	Low-Medium (similar to Deepfakes)

GAN-based methods (Deepfakes, FaceShifter, DeepFakeDetection) are detected with higher confidence by Stream 3 because generative adversarial training induces characteristic frequency-domain fingerprints, periodic artefacts in the high-frequency DCT components documented by Frank et al. (2020) and Durall et al. (2020). The FFT magnitude map exposes these patterns even when spatial appearance is convincing. Stream 1 contributes additional signal through semantic inconsistency (identity leakage from the source face).

Expression-transfer methods (Face2Face, NeuralTextures) produce the most subtle spatial artefacts because they preserve the target identity and change only the expression or texture. Stream 2 (periocular crop) is most discriminative here: the mouth region targeted by Face2Face consistently shows boundary artefacts where the synthesised expression meets the original skin texture, and NeuralTextures produces visible rendering seams in the eye region that the 224×224 periocular crop is positioned to capture at sufficient spatial resolution.

Identity-swap with occlusion (FaceShifter) is the hardest single method because it is specifically designed to handle challenging cases (glasses, hair occlusions) that defeat earlier swap methods. Nevertheless, Stream 1's pretrained identity features distinguish the source identity from the target face structure, and residual periocular inconsistencies remain detectable by Stream 2.

Section 13

Key Findings

- **1. Multi-stream fusion is essential but marginal gains depend on stream quality.**

The full three-stream model (E11, 96.6%) outperforms every single-stream and dual-stream variant. However, Stream 1 alone (95.1%) already achieves competitive performance, and Stream 3 in isolation (71.9%) is the weakest contributor. Fusion gains are real (+1.5pp) but require careful stream design.

- **2. Transfer learning dominates deepfake detection at this scale.**

Stream 1 (pretrained ResNet-18, ~11.2M params) achieves 95.1% accuracy. Stream 2 (trained from scratch, ~1.3M params) achieves 85.7%. The 10.6pp gap confirms that ImageNet pretraining provides features that generalise strongly to face manipulation detection, even across the large domain gap from natural images to faces.

- **3. Dropout tuning matters more than architecture complexity.**

Reducing dropout from 0.5 to 0.3 (E11 vs E09) yields +1.10pp, a larger gain than adding SE attention (+0.14pp). The network is in a high-bias regime at dropout=0.5; relaxing regularisation unlocks the model's representational capacity.

- **4. Adam with cosine annealing is the correct optimiser for multi-stream networks.**

Adam (95.5%) outperforms SGD (92.8%) by 2.7pp. Heterogeneous gradient scales from three architecturally diverse streams demand per-parameter adaptive rates.

- **5. ELU is the best activation for the custom CNN stream.**

ELU: 95.67% vs ReLU: 94.57% vs Leaky ReLU: 94.29%. ELU's smooth negative half-plane preserves gradient flow for near-zero activation values, critical for detecting subtle periocular manipulation artefacts.

- **6. Batch Normalisation in Stream 2 is redundant in the full model.**

Removing Stream 2 BN: Acc 94.95% vs With BN: 94.57% ($\Delta = +0.38\text{pp}$). The fusion head's BN1d already normalises the full 896-d vector. Removing per-stream BN saves 31% training time with no accuracy cost.

- **7. SE channel attention provides modest but consistent benefit.**

E11 (with attention): 96.57% vs E12 (without): 96.43% ($\Delta \text{ Acc} = +0.14\text{pp}$, $\Delta \text{ AUC} = -0.002$). Attention adds marginal accuracy at a near-zero parameter cost; its primary value may be interpretability (stream importance weights).

Section 14

Real-World Reflection

Deployment Challenges in Pakistan

Pakistan presents a uniquely challenging deployment environment for deepfake detection systems. Three infrastructure constraints are particularly acute:

■ **Low-resource devices:** The majority of Pakistani internet users access content on entry-level Android smartphones with 2–4 GB RAM and no dedicated GPU. MultiCue-DF's 12.4M parameters require ~50 MB of memory in float32, which is feasible for inference on device, but the three-stream preprocessing pipeline (full-face crop, periocular crop, FFT computation) adds latency that may not be acceptable for real-time social media scanning. Model distillation into a MobileNet-scale architecture is a necessary next step for on-device deployment.

■ **Compressed WhatsApp videos (C40-equivalent quality):** Content shared on WhatsApp undergoes aggressive re-encoding that introduces compression artefacts at a level comparable to FaceForensics++ C40. Our C23-trained model may lose significant accuracy on heavily compressed content because compression smooths the frequency-domain fingerprints that Stream 3 relies upon. Fine-tuning on a mixture of C23 and C40 data, or using compression-robust features, is essential for Pakistan-specific deployment.

■ **Urdu/Pashto language metadata:** Contextual signals (captions, hashtags, surrounding text) that could aid detection are predominantly in Urdu or Pashto written in Nastaliq or Latin scripts. Existing NLP models for South Asian languages are significantly less mature than English counterparts, making multimodal detection (combining video and text signals) harder to implement. Pakistan-specific research investment in low-resource NLP is needed to enable this.

Ethical Considerations

■ **False positives harm innocent individuals:** A deepfake detector that wrongly labels genuine content as manipulated can suppress legitimate speech, damage reputations, and, in Pakistan's political environment, be weaponised to discredit real footage of political events. At 96.57% accuracy, approximately 3.4% of real videos are misclassified as fake on our test set. In a high-stakes political context, this false positive rate is unacceptably high for automated removal. Any deployed system must present confidence scores rather than binary verdicts, and route borderline cases to human reviewers.

■ **Dual-use concern:** Publishing the specific architecture and failure modes of a deepfake detector, as this report does, creates a roadmap for adversarial circumvention. A sufficiently motivated adversary can use knowledge of the three-stream architecture to design GAN-based fakes that suppress both spatial and frequency artefacts simultaneously. This dual-use tension is fundamental to the field: detection and generation are in an adversarial arms race. Responsible disclosure practices, like sharing architecture details with trusted stakeholders while withholding operational deployment details, partially mitigate this risk.

■ **Governance frameworks are as important as technology:** Technical detection alone cannot solve the deepfake problem. Legal frameworks that criminalise malicious deepfakes, platform policies that mandate labelling of AI-generated content, and media literacy education that trains citizens to be critical consumers of digital media are complementary and arguably more impactful interventions than detection accuracy improvements.

Recommended Integration Pathways

- Browser extension for news platforms (Dawn, Geo, ARY News): intercepts video content, runs lightweight detection, overlays a confidence badge. Similar to the NewsGuard extension model.
- WhatsApp Business API integration: media shared in public groups could be queued through a server-side detection pipeline that appends a 'potential deepfake' metadata flag before delivery. WhatsApp already uses this pattern for spam detection.
- Government media monitoring: PEMRA (Pakistan Electronic Media Regulatory Authority) could integrate detection into their broadcast monitoring infrastructure for real-time flagging of potentially manipulated political video content.

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Limitations

- Single dataset: all experiments use FaceForensics++ C23. Cross-dataset generalisation to Celeb-DF, DFDC, or in-the-wild deepfakes is not evaluated.
- Fixed frame sampling: 10 uniformly sampled frames per video ignores temporal dynamics. Temporal inconsistencies (eye blink frequency, audio-lip sync) are unused signals.
- No per-manipulation-method accuracy breakdown: the model is evaluated on a mixed test set. Some methods (e.g., NeuralTextures) are likely harder to detect than others, but this is not quantified.
- Stream 3 in isolation is weak (71.9% accuracy): the FFT stream's low standalone performance suggests it may be learning noise correlations rather than genuine frequency fingerprints, particularly under C23 compression.
- Manual hyperparameter search: learning rate, weight decay, and architecture depth are chosen by hand. Bayesian optimisation could find better configurations.
- Adversarial robustness is untested: the model is trained and tested on standard (non-adversarial) samples. Perturbation-based evasion attacks are not evaluated.
- No real-time inference benchmark: latency and throughput on target deployment hardware (mid-range Android) are not measured.

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Future Work

- Temporal modelling: replace frame-level classification with a lightweight transformer or LSTM over the 10-frame sequence to capture motion inconsistencies and audio-lip sync errors.
- Cross-dataset evaluation: test on Celeb-DF v2, DFDC, and a custom South Asian faces dataset to measure out-of-distribution generalisation.
- Combined ablation (ELU + no BN): run a single experiment combining the two best findings from the activation and BN analyses. Expected accuracy: >96.8%.
- Gated fusion for Stream 3: replace simple concatenation with a learned gate that suppresses Stream 3's contribution when FFT confidence is low, addressing the counter-intuitive finding that the full model underperforms Stream 1+2.
- Knowledge distillation: distil the 12.4M-param teacher into a MobileNetV3-scale student (<5M params) for on-device inference on Pakistani low-end smartphones.
- Grad-CAM visualisation: generate per-stream saliency maps to visualise which facial regions and frequency bands drive each prediction, improving interpretability.
- Adversarial training: augment training data with adversarially perturbed samples to improve robustness against evasion attacks.
- Multimodal detection: incorporate voice deepfake detection (e.g., AASIST model) alongside video to provide richer and more robust evidence for combined audio-visual deepfakes.

Section 17

Conclusion

This report presents MultiCue-DF, a three-stream deepfake detection framework that fuses spatial, periocular, and frequency-domain evidence via Squeeze-and-Excitation channel attention. Evaluated on FaceForensics++ C23 through twelve controlled ablation experiments, the best configuration (**exp11_full_model_dropout03**) achieves **96.57% accuracy, AUC 0.9904, F1 0.9799** on the held-out test set, competitive with published state-of-the-art results on this benchmark.

The ablation study establishes seven clear findings: (1) multi-stream fusion outperforms all single-stream baselines; (2) pretrained ResNet-18 features dominate detection at this scale; (3) dropout tuning (0.5→0.3) yields larger gains than architectural complexity; (4) Adam with cosine annealing is essential for multi-stream optimisation; (5) ELU activation improves accuracy and training speed in the custom CNN stream; (6) Batch Normalisation in Stream 2 is redundant and its removal saves 31% training time; and (7) SE channel attention provides modest but consistent benefit.

Beyond technical performance, this work reflects on the urgent real-world context for deepfake detection in Pakistan, a country where political disinformation spreads at viral speed on low-bandwidth mobile networks. Responsible deployment requires not only high accuracy but also hardware-efficient models, human-in-the-loop review for borderline cases, and complementary governance frameworks. We hope this work contributes both technical insights and a grounded perspective on the sociotechnical dimensions of AI-driven media forensics.

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